

SOFTWARE

ENGINEERING

ASSIGNMENT - I

714024205109

C. Nivetha

IT - 'B'

1. A startup wants to develop a food delivery application. The client expects the application to be completed in six months, with the possibility of adding new features later.

i) Which software process model would you recommend?

Incremental Process Model

ii) Why is this model suitable?

The incremental model develops software in small modules. Each increment is designed, coded, tested & delivered independently. Since the client wants a working application within 6 months & expects additional features later, it is the best choice.

The first version can include:

- User Registration / login
- Restaurant list
- Food ordering

The next versions can add:

- Online Payments
- Rating & Reviews
- Live order tracking
- Coupons & Offers

iii) mention 3 advantages of your chosen model.

1. Working software is delivered early
2. New features can be added easily.
3. Easy to test & debug

2. Company A develops software using the waterfall model, while company B uses the incremental model.

i) Compare both models.

waterfall model	Incremental model
Linear sequential model	Software built in small increments
Requirements fixed at the beginning	Requirements can change between increments
Testing after development	Testing in every increment
Difficult to make changes	Easy to make changes
High risk	Lower risk
Slow delivery	Faster delivery

ii) Which company can deliver software to customers earlier?
Company B using the Incremental model.

iii) Justify your answer

The incremental model delivers a working version after every increment. Customers can start using important features while the remaining features are under development. In contrast, the waterfall model delivers the complete software only after all phases are completed.

3. Your college wants to develop an Online Exam System.

Task:

Prepare a software development plan by:

i) Identifying the software process:

Software Development Life cycle (SDLC)

ii) Choosing an appropriate process model:

Incremental model.

iii) Drawing its basic phases

Requirement Analysis



System Design



Implementation



Testing



Deployment



Maintenance.

iv) Why is it suitable

→ Features such as login, student registration, exams & result can be developed 1 by one.

→ Easy to update & maintain

→ Customer feedback can be incorporated after

each release.

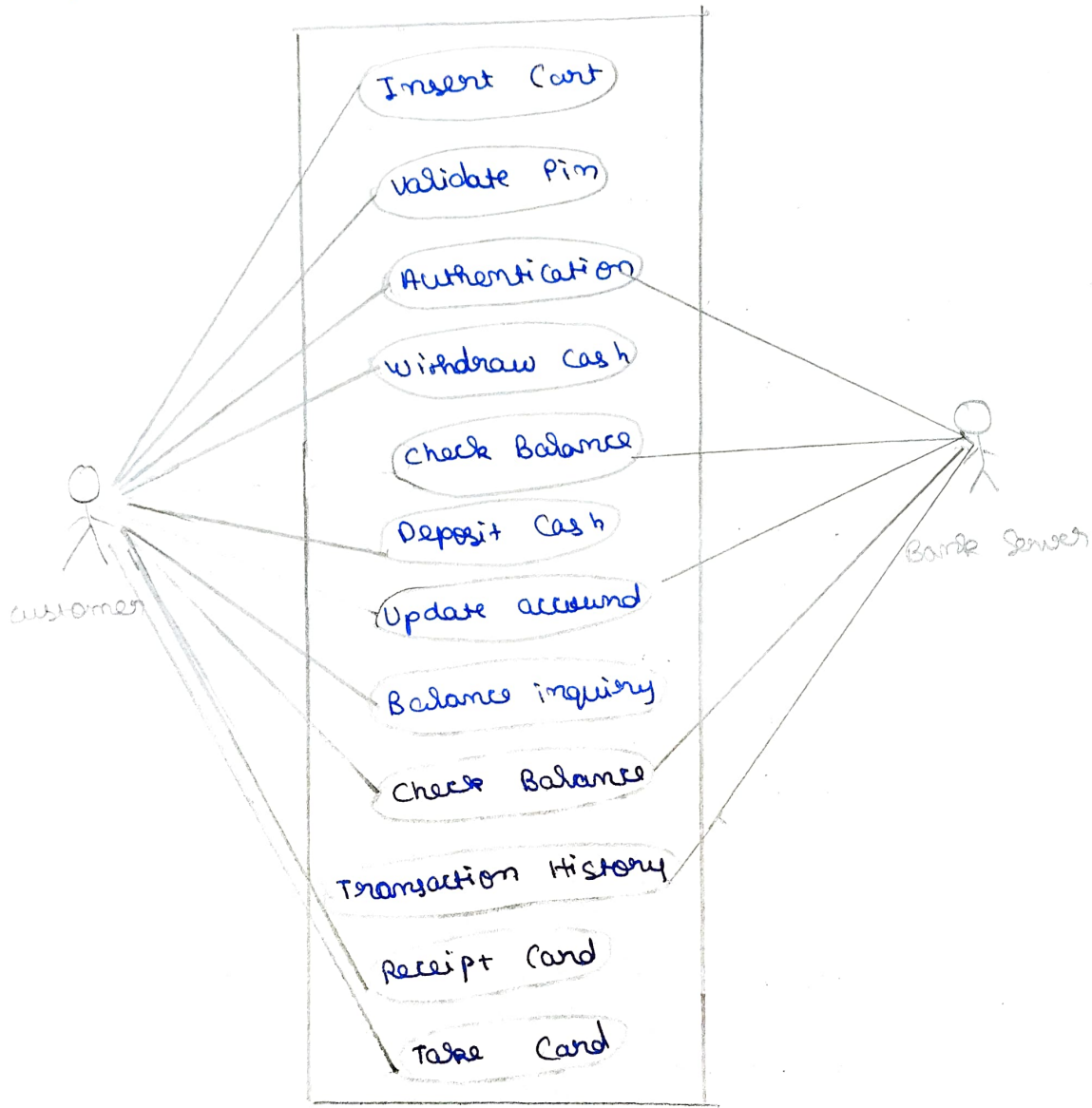
4. Develop a use case Diagram for an ATM System

Include:

→ Actors

→ Use cases

→ Relationships



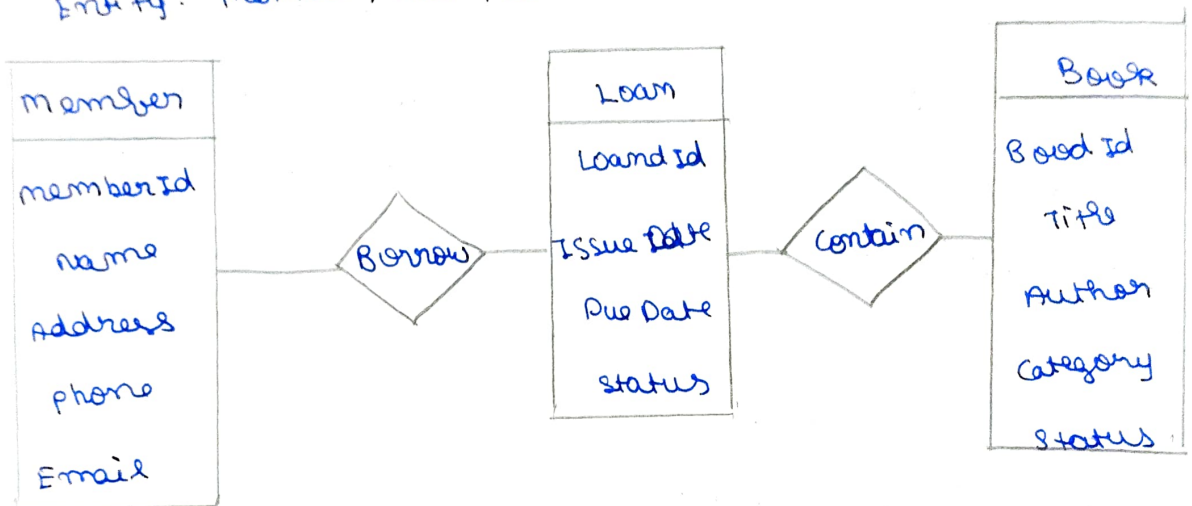
4. Design & Analysis model for a library management system
Include:

1. Data Model
2. Class Model
3. Scenario Based Model
4. Behavioral Model

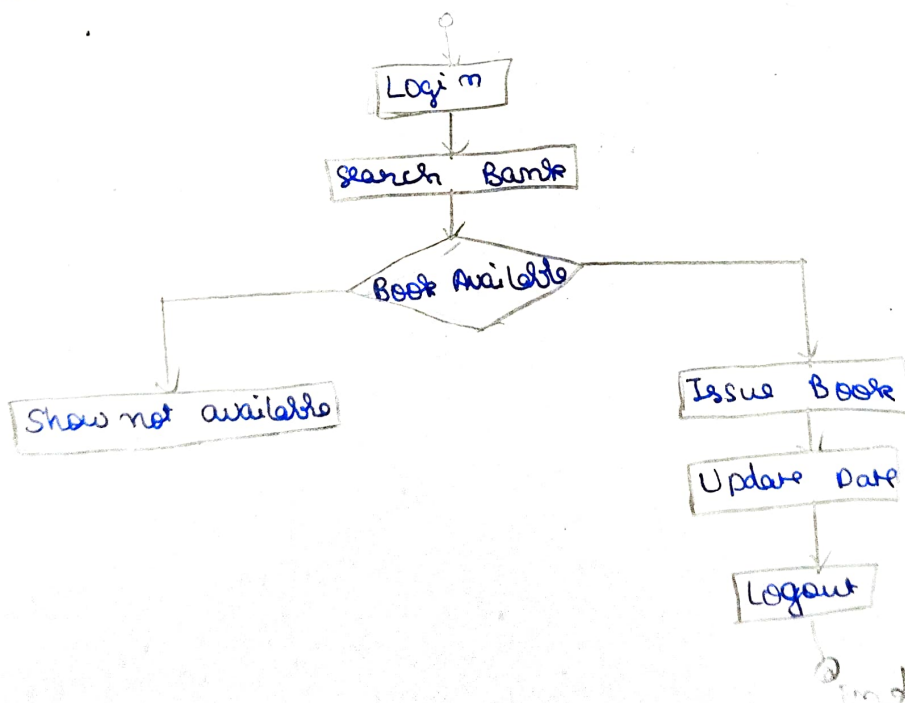
Present your design to the class

1. Data Model

Entity: member, loan, book



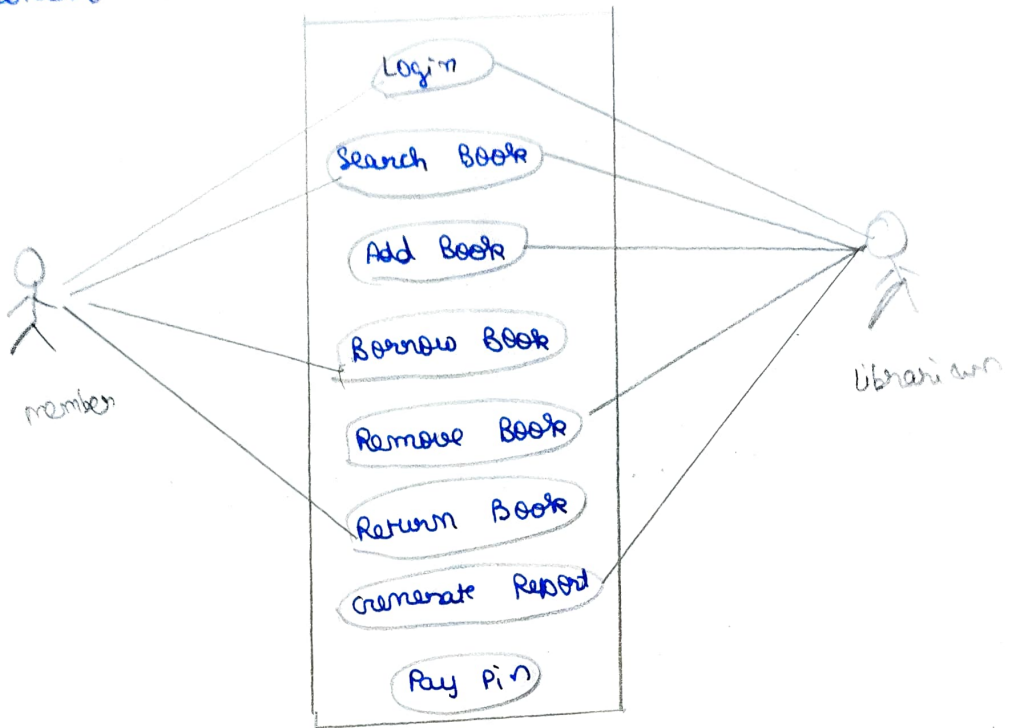
4. Behavioural Model:



2. Class model

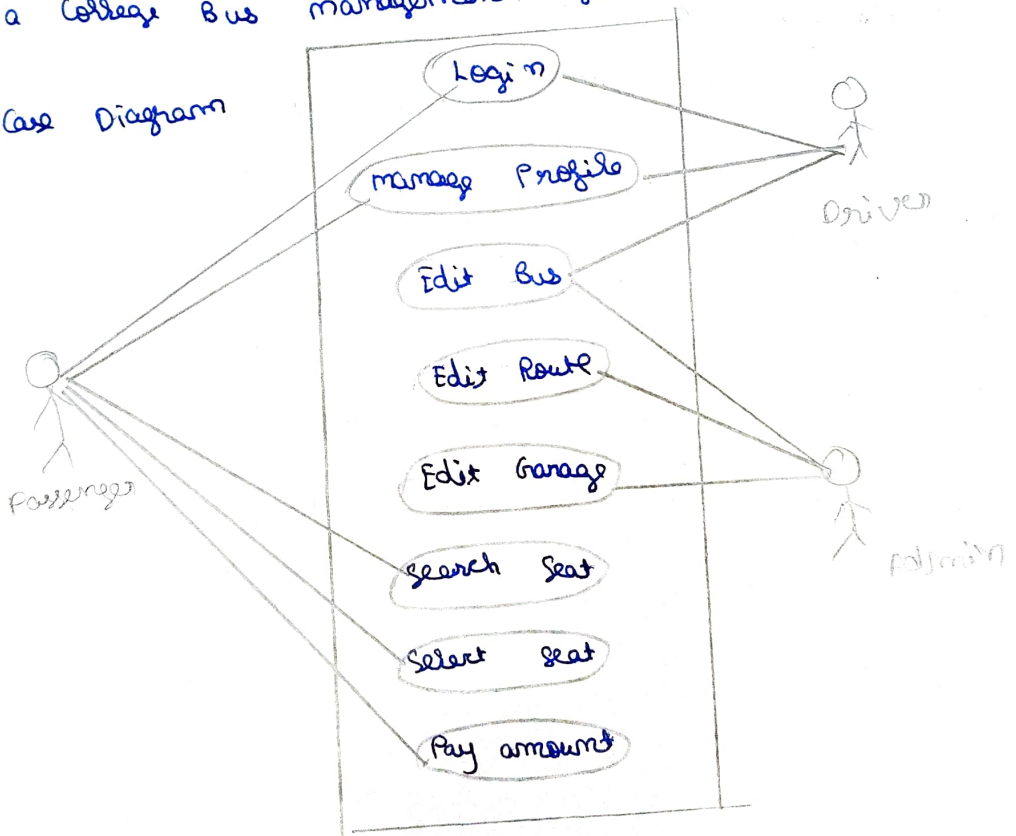


3 Scenario based model



5. Design a College Bus management System.

1. Use Case Diagram



2. Activity Diagram

